

Discipline	Category	Subcategory	Element	Attributes	Notes
Mechanical Equipment	Pumps	Centrifugal	Base pump unit	Design flowrate, Head, NPSH, Speed, Materials, Seal type	Include datasheet, vendor, tag
Mechanical Equipment	Pumps	Positive Displacement	Gear/Diaphragm	Displacement, Max pressure, Viscosity limits, Materials	
Mechanical Equipment	Vessels	Pressure Vessel	Reactors/Separators	Design pressure/temp, MAWP, materials, relief device	
Mechanical Equipment	Tanks	Storage Tank	Fixed/Float Roof	Capacity, material, coating, bunding/containment, height, width, aspect ratio	
Mechanical Equipment	Heat Exchangers	Shell-and-tube	Shell-tube unit	Heat duty, LMTD, materials, fouling factor, U-value	
Mechanical Equipment	Compressors/Blowers	Centrifugal/Turbo	Compressors	Flow, pressure ratio, driver type, lubrication, interstage cooling	
Mechanical Equipment	Material Handling	Conveyors/Hoists	Belt/ Screw	Capacity, speed, drive, guards, throughput	
Mechanical Equipment	Utilities	Cooling Tower	Open/Closed	Cooling duty, cycles of concentration, blowdown	
Mechanical Equipment	Specialty	Agitators	Top/Side Entry	Power, tip speed, seal type, shaft material	
Mechanical Equipment	Process Equipment	Grinding Mill	Ball/Rod/Mill	Power, throughput, liner material, grinding media	
Civil	Site Works	Grading	Cut/fill	Finished grade, slopes, erosion control, topsoil	
Civil	Access	Roads/Paving	Site roads	Pavement type, width, drainage, loading	
Civil	Stormwater	Retention	Basins/Channels	Detention volume, outlet controls, SWPPP	
Civil	Utilities - Site	Duct banks	Electrical ductbank	Conduit size, cover depth, pull chambers	
Civil	Hydrology	Culverts/Drainage	Culverts	Size, invert, headwall, scour protection	
Civil	Survey	Topography	Benchmarking	Datum, coordinate system, control points	
Structural	Steelwork	Platforms/Walkways	Access platform	Loading criteria, handrails, gratings, access points	
Structural	Pipe Support	Racks & Supports	Pipe rack	Span, load rating, thermal movement allowances	
Structural	Buildings	Enclosures	Control room / MCC building	Occupancy type, fire rating, service penetrations	
Structural	Foundations Interface	Anchor/Embed Plates	Embed plate	Plate size, anchor bolt layout, concrete strength	
Concrete	Foundations	Footings	Spread footings	Concrete class, reinforcement, bearing capacity	
Concrete	Slabs	Slab on grade	Equipment plinths	Thickness, joints, surface finish, load	
Concrete	Retaining	Retaining walls	Gravity/Sheet pile	Height, surcharge, drainage, waterproofing	
Concrete	Precast	Manholes/Precast	Manhole chamber	Design load, gasket, access cover	
Piping	Process Lines	Large Bore	Slurry / Main headers	Diameter, schedule, lining/lining material, expansion allowances	

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Piping	Process Lines	Small Bore	Instrument & Utility lines (<2")	Material spec, isolation valves, tagging, support spacing	
Piping	Materials	Carbon steel / SS / HDPE / FRP	Material selection	Specification (ASTM/ASME), corrosion allowance, coating	
Piping	Insulation	Heat tracing	Insulation & HT	Insulation type, thickness, heat tracing circuits, control	
Piping	Testing	Hydrotest & Pneumatic	Pressure testing	Test pressure, acceptance criteria, duration	
Instrumentation	Flow / Level	Flow meters	Coriolis/Ultrasonic	Accuracy, range, mounting, straight-run requirements	
Instrumentation	Temperature & Pressure	Transmitters	Temp & Pressure TX	Range, accuracy, material, isolation	
Instrumentation	Analyzers & Safety	Gas detectors, pH	Analyzers	Calibration, sample conditioning, response time	
Instrumentation	Cabling & Termination	Junction Boxes	Field junctions	Cable types, gland sizes, NEMA/Ingress rating	
Control	Control Systems	PLC / DCS / SCADA	Control platform	I/O count, redundancy, vendor, programming language	
Control	Interfaces	HMI / Historian	Operator stations	Screens, alarm philosophy, trend requirements	
Control	Network & Cyber	Network switches / Firewalls	Industrial network	Architecture, VLANs, cybersecurity controls	
Control	I/O	Remote I/O panels	Field I/O	Location, IP/analog, termination, power supply	
Electrical	Distribution	Transformers	Step-down/step-up	kVA rating, cooling type, grounding	
Electrical	Switchgear	MCC / SWGR	Motor control center	AIC rating, upstream protection, TB layouts	
Electrical	Cabling	Power & Control cable	Tray and routing	Tray fill, segregation, derating	
Safety/Environment	Fire Protection	Sprinkler / Hydrants	FP systems	Design density, hydrant spacing, deluge zones	
Safety/Environment	Containment	Spill bunding	Bunds	Capacity, drainage, liners	
Safety/Environment	Ventilation	Fume / Dust extraction	Ducting & fans	Air changes, filtration, duct sizing	
Utilities	Compressed Air	Air network	Compressors & Piping	Pressure, dewpoint, receiver sizing	
Utilities	Cooling Water	CW network	Pumps & Heat exchangers	Flow, temperature, water treatment requirements	
Construction/Commissioning	QA/QC	Inspections & Testing	Quality plan	Hold points, test records, certificates	
Construction/Commissioning	Commissioning	Pre-startup Safety Reviews	Commissioning plan	Loop checks, SAT, punchlists	
Cost	Capex	Direct/ Indirect			

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	Contingency				